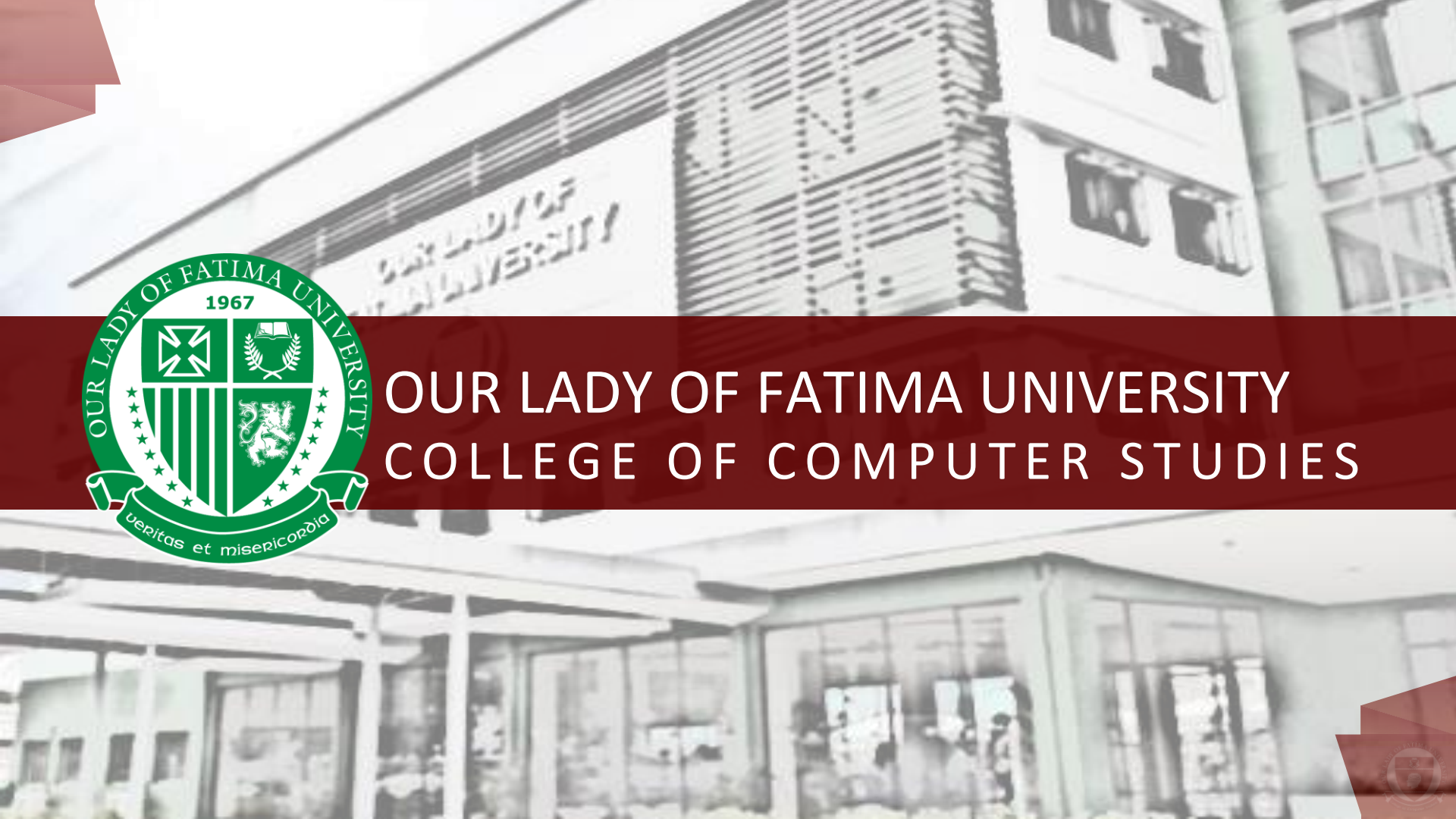
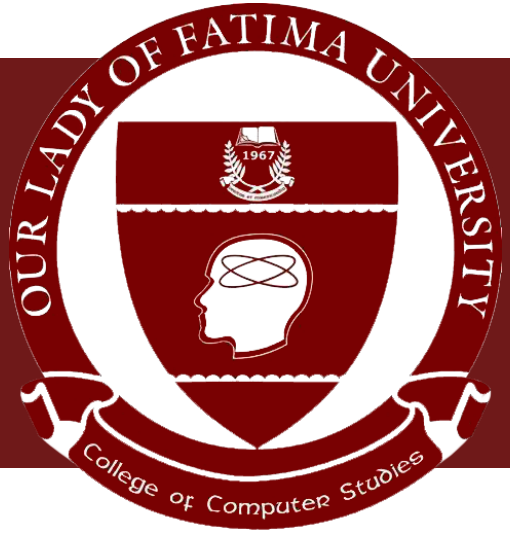
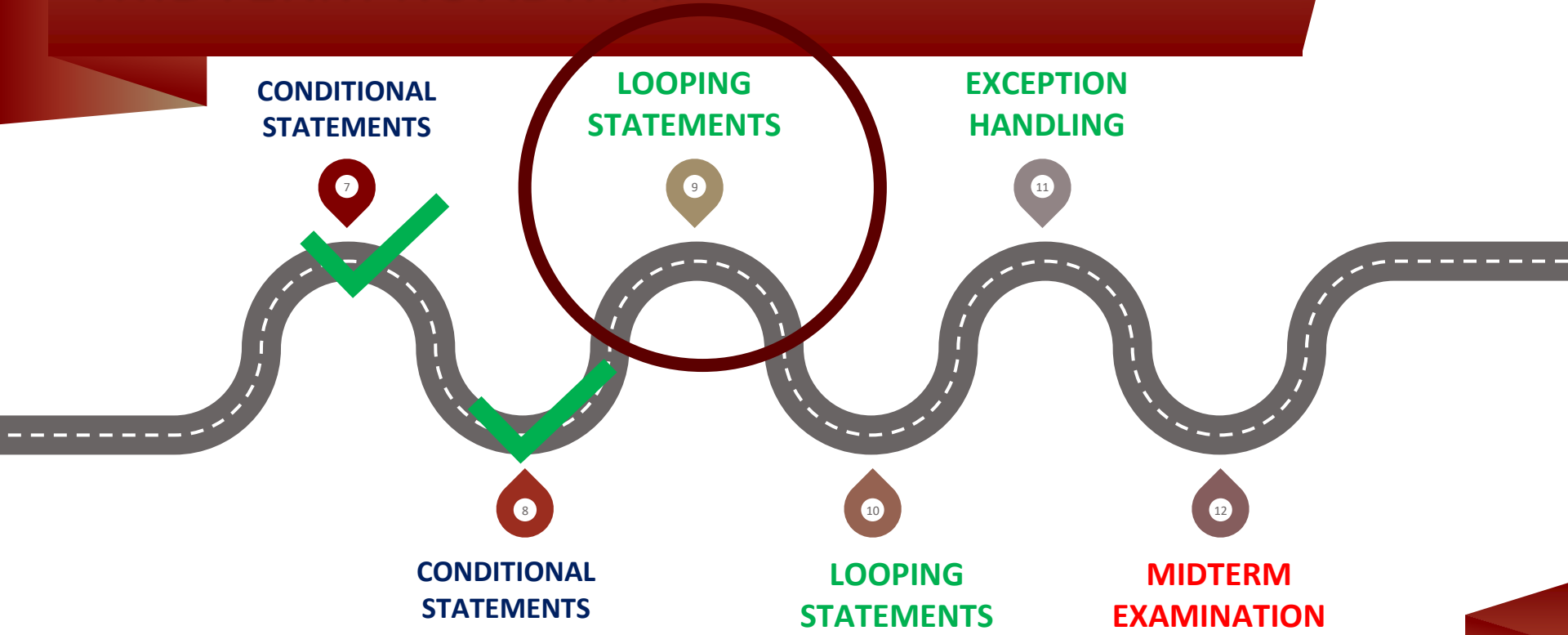


OUR LADY OF FATIMA UNIVERSITY
COLLEGE OF COMPUTER STUDIES



OOPR211
OBJECT-ORIENTED
PROGRAMMING I

MIDTERM ROADMAP



Module 6: JAVA CONTROL STATEMENTS – LOOPING STATEMENTS

Module Objectives:

1

Know the syntax of the repetition structure.

2

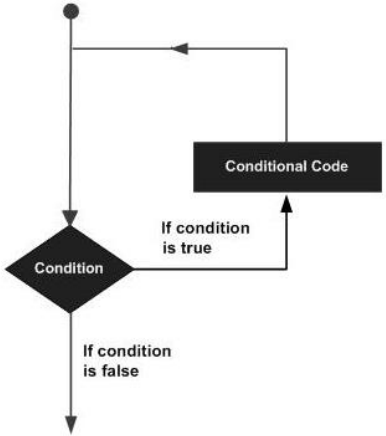
Practice Problem solving solution using a different approach like looping statement.

Topics:

- For Loop
- While Loop
- Do…while Loop

LOOPING STATEMENTS

LOOP STATEMENT



ALLOWS US TO EXECUTE A STATEMENT OR GROUP OF STATEMENTS MULTIPLE TIMES AND FOLLOWING IS THE GENERAL FORM OF A LOOP STATEMENT IN MOST OF THE PROGRAMMING LANGUAGES.

CATEGORIES

for loop	• Execute a sequence of statements multiple times and abbreviates the code that manages the loop variable.
while loop	• Repeats a statement or group of statements while a given condition is true. It tests the condition before executing the loop body.
do...while loop	• Like a while statement, except that it tests the condition at the end of the loop body.

1 Understanding *for loop*

IT IS USEFUL WHEN YOU KNOW HOW MANY TIMES A TASK IS TO BE REPEATED.

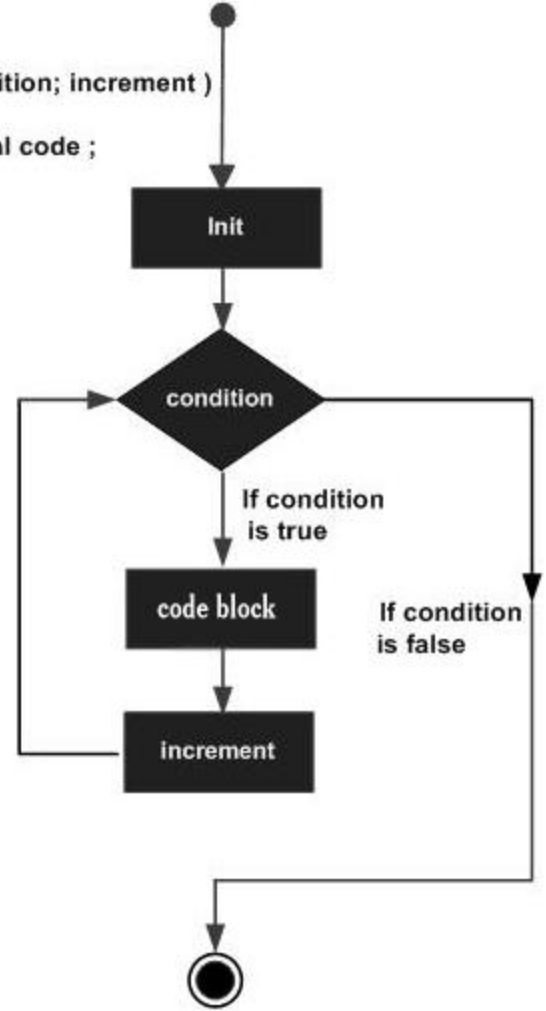
for LOOP

A REPETITION CONTROL STRUCTURE THAT ALLOWS YOU TO EFFICIENTLY WRITE A LOOP THAT NEEDS TO BE EXECUTED A SPECIFIC NUMBER OF TIMES.

for LOOP STATEMENT

```
for(initialization; Boolean_expression; update) {
   // Statements
}
```

```
for( init; condition; increment )
{
    conditional code ;
}
```



for LOOP DEMO

Code

Main.Java

```java
import java.util.Scanner;

class Main {
    public static void main(String args[]) {
        Scanner input = new Scanner(System.in);

        int n = input.nextInt();
        int sum = 0;

        for(int iter = 1; iter <= n; iter++)
            sum = sum + iter;

        System.out.println("The sum from 1 to " + n + " is " + sum + ".");
    }
}
```

OUTPUT

Code

Main.Java

```java
import java.util.Scanner;

class Main {
    public static void main(String args[]) {
        Scanner input = new Scanner(System.in);

        int n = input.nextInt();
        int sum = 0;

        for(int iter = 1; iter <= n; iter++)
            sum = sum + iter;

        System.out.println("The sum from 1 to " + n + " is " + sum + ".");
    }
}
```

Custom Input

```
7
```

Output

```
The sum from 1 to 7 is 28.
```

2

Understanding *while loop*

while LOOP

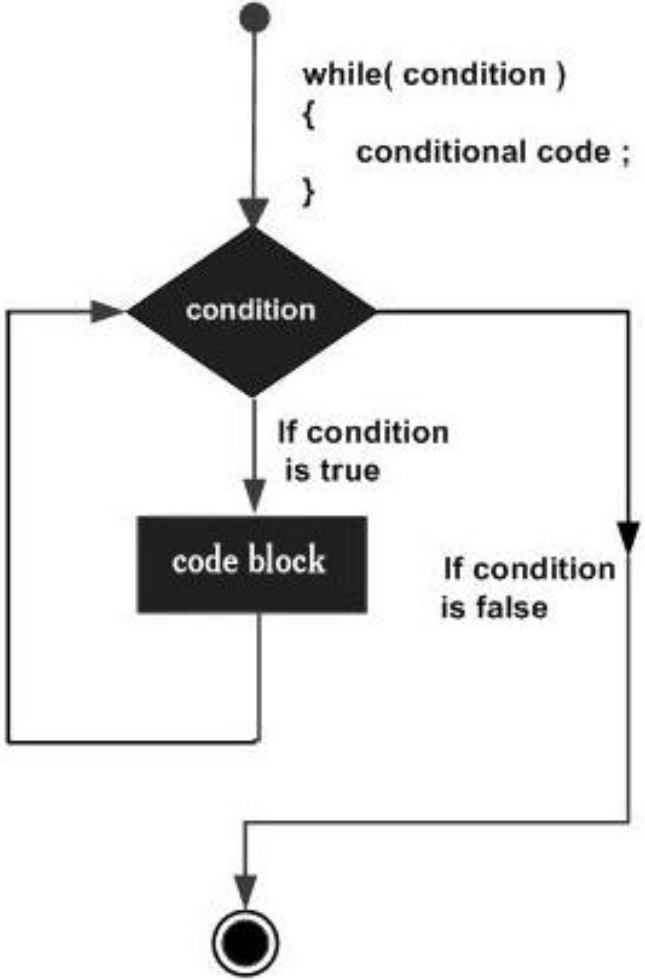
A WHILE LOOP STATEMENT IN JAVA PROGRAMMING LANGUAGE REPEATEDLY EXECUTES A TARGET STATEMENT AS LONG AS A GIVEN CONDITION IS TRUE.

while LOOP STATEMENT

```
while(Boolean_expression) {
// Statements
}
```

while

- Here, key point of the *while* loop is that the loop might not ever run. When the expression is tested and the result is false, the loop body will be skipped and the first statement after the while loop will be executed.



while LOOP DEMO

Code

Main.Java

```java
import java.util.Scanner;

class Main {
    public static void main(String args[]) {
        int n = 0, iter = 1, sum = 0;

        Scanner input = new Scanner(System.in);

        n = input.nextInt();

        while(iter <= n) {
            sum = sum + iter ;
            iter++;
        }

        System.out.println("The sum from 1 to " + n + " is " + sum + ".");
    }
```

OUTPUT

Code

Main.Java

```java
import java.util.Scanner;

class Main {
    public static void main(String args[]) {
        int n = 0, iter = 1, sum = 0;

        Scanner input = new Scanner(System.in);

        n = input.nextInt();

        while(iter <= n) {
            sum = sum + iter ;
            iter++;
        }

        System.out.println("The sum from 1 to " + n + " is " + sum + ".");
    }
```

Custom Input

```
5
```

Output

```
The sum from 1 to 5 is 15.
```

3

Understanding do…*while loop*

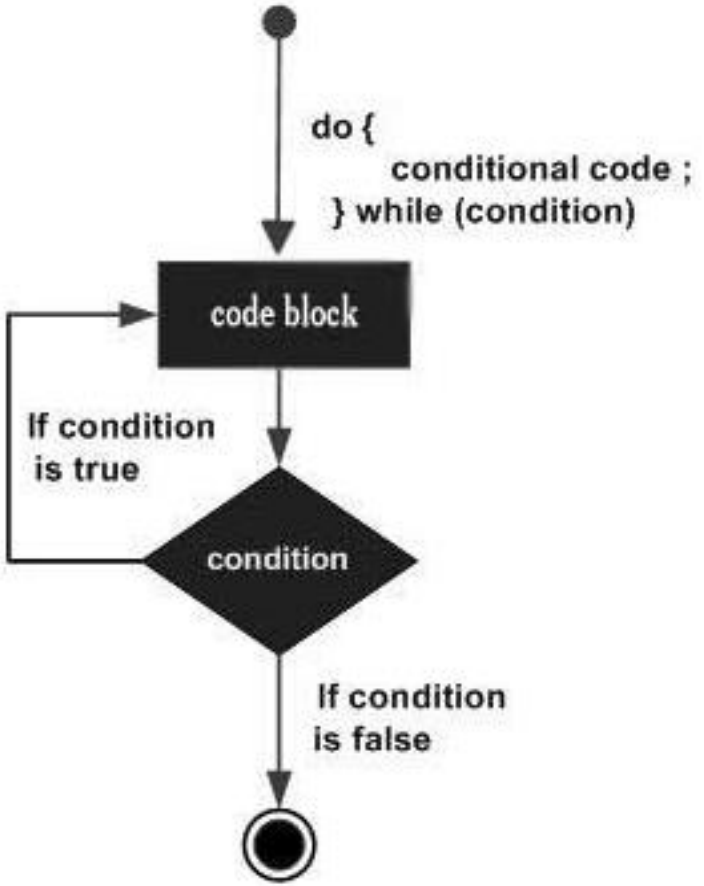
do...while LOOP

A DO...WHILE LOOP IS SIMILAR TO A WHILE LOOP, EXCEPT THAT A DO...WHILE LOOP IS GUARANTEED TO EXECUTE AT LEAST ONE TIME.

do…while LOOP STATEMENT

```
do {
// Statements
}while(Boolean_expression);
```

do-while


do {
conditional code ;
} while (condition)
code block
If condition
is true
condition
If condition
is false

do...while LOOP DEMO

```
public class Test {

   public static void main(String args[]) {
      int x = 10;

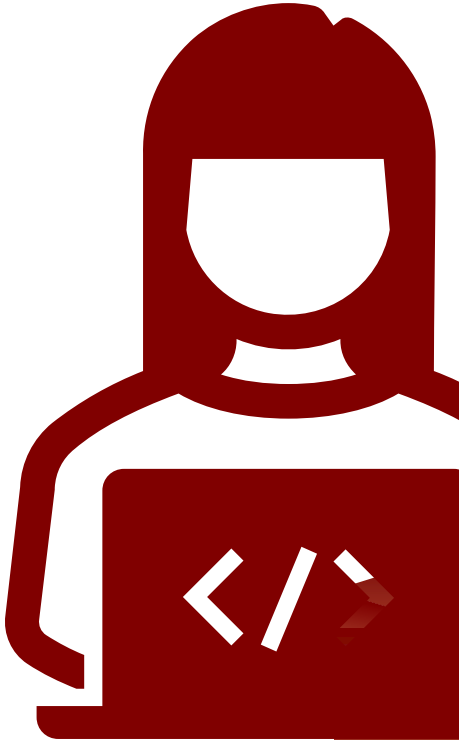
      do {
         System.out.print("value of x : " + x );
         x++;
         System.out.print("\n");
      }while( x < 20 );
   }
}
```

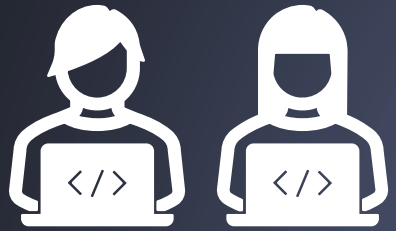
OUTPUT

```
value of x : 10
value of x : 11
value of x : 12
value of x : 13
value of x : 14
value of x : 15
value of x : 16
value of x : 17
value of x : 18
value of x : 19
```

EXERCISES:

1. Test if the value of the variable count is greater than 10. If it is, print “Count is greater than 10”, otherwise, print “Count is less than 10”.
2. Test if the value of the variable number is an odd number. If it is, print “This number is not an even number” otherwise, print “This number is an odd number.”
3. Test if the value of the variable age is greater than 18 and less than or equal to 40. If it is, print “You’re not getting old!” otherwise, print “You’re not getting younger.”





Check your Understanding

Write a program to output summation of a set of numbers.

Example Output:

Please input how many numbers you would like to add: 4

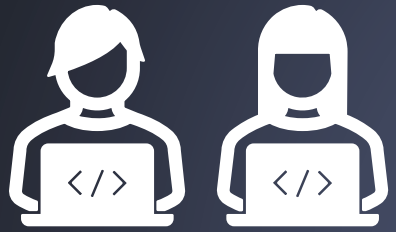
Please enter a value: 3

Please enter a value: 1

Please enter a value: 2

Please enter a value: 3

The sum of the 4 values is 9



Check your Understanding

Write a program which should be able to reproduce the following example output:

```
*
***
*****
*******
*********
```

QUESTIONS?

Thank you!

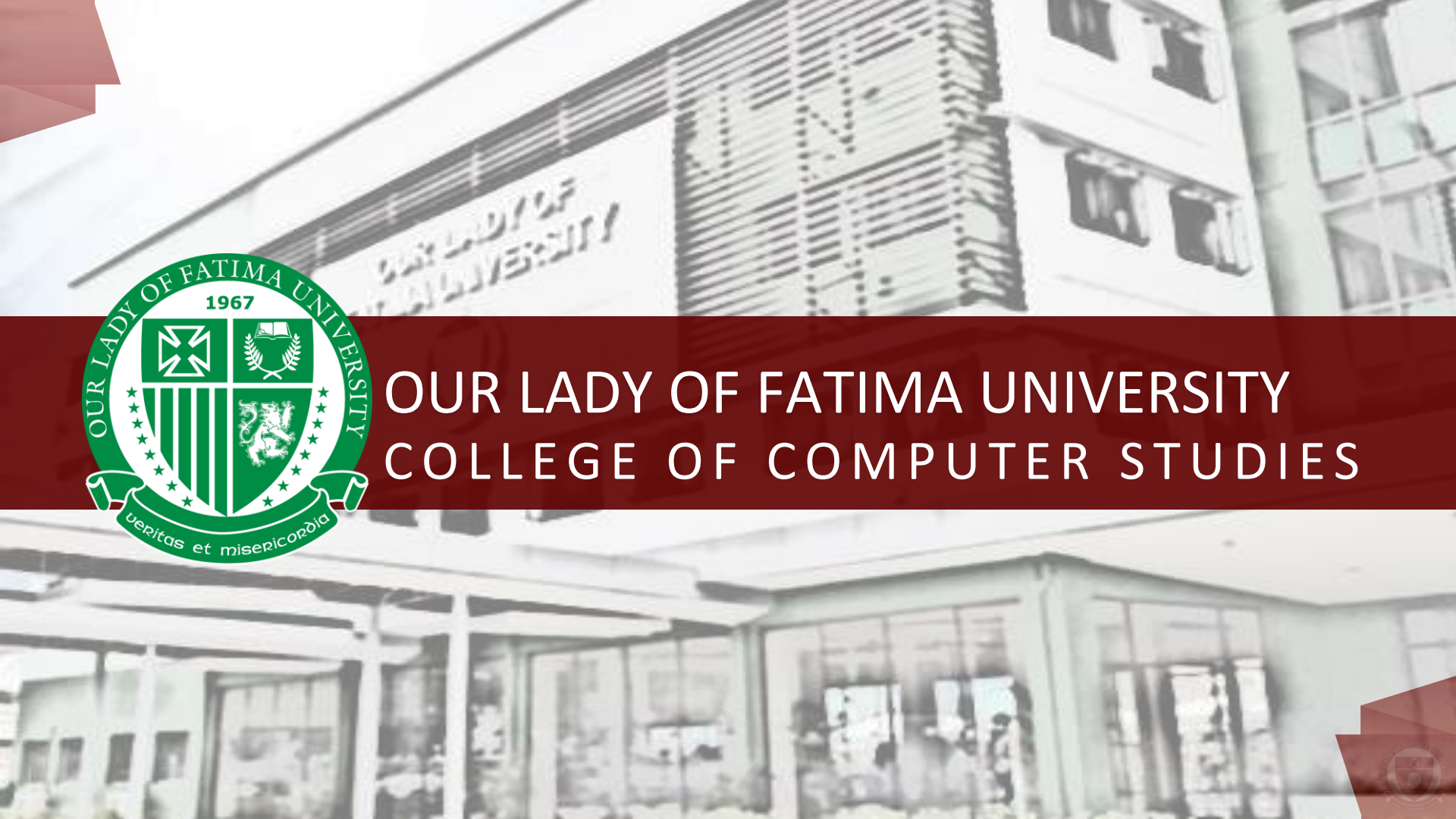
Credits

Special thanks to all the people who made and released these awesome resources for free:

- Presentation template by SlidesCarnival
- Photographs by Unsplash

Additional References:

- https://www.edureka.co/blog/what-is-java/
- http://ecomputernotes.com/java/what-is-java-language/types-of-java-programs
- https://app.codechum.com/teacher/classes/504/timeline/lessons/java-loops/for-loop/



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